
THE IDENTITY CRISIS

Surrogate vs. Natural Keys: Building Stable Foundation for BI

ORACLE DATA WAREHOUSING BEST PRACTICES

THE FOUNDATION

Why Business Keys are not Persistent Identifiers

| DEFINING THE KEY CONFLICT

NATURAL KEYS

Identities defined by business processes (SKUs, Emails, Serial Numbers). They are meaningful but highly volatile.

SURROGATE KEYS

Artificial, warehouse-generated identifiers. They are anonymous, immutable, and strictly integer-based for maximum performance.



| SOURCES OF KEY VOLATILITY



SYSTEMS MERGE

Collision of duplicate natural keys (e.g., both merging companies have a Product ID '101').



RECYCLING

Operational systems reuse SKUs for new products after an old product is discontinued.



MIGRATION

Customer changes their email or username, breaking the historical natural key linkage.

FEATURE COMPARISON MATRIX

Feature	Natural Key	Surrogate Key
Origin	Operational Source System	Data Warehouse ETL Process
Business Meaning	High (Readable)	None (Artificial)
Stability	Volatile (Changes Over Time)	Immutable (Never Changes)
Join Performance	Slower (Variable String)	Optimal (Fixed Integer)

| SCD TYPE 2 LIFECYCLE DEMO



| JOIN PERFORMANCE BENCHMARK



*Benchmark based on 100M row Fact Table Join in Oracle Autonomous DB.

| PRESERVING HISTORY

Surrogate keys serve as a "Time Machine" for your data.

By decoupling the identity from the business key, the warehouse preserves exactly what was true at the point of transaction.

- Historical Auditability
- Referential Integrity
- Logical Separation



"Business keys change. Systems merge. SKUs recycle. To maintain a truly immutable historical record, Surrogate Keys are not optional—they are foundational."

— Oracle Data Warehousing Best Practices Guide

| GLOBAL USE CASES



PRODUCT TRACKING

Decoupling seasonal re-packaging from SKU IDs.



CUSTOMER IDENTITY

Handling email migrations without losing history.



REGIONAL MERGE

Consolidating diverse ID schemas during M&A.

| THE ROI OF STABILITY

75%

Faster Joins

0%

Data Overwrites

100%

Historical Clarity

QUESTIONS?

Mastering the Transition from Natural to Surrogate Keys

www.oracle.com/data-warehouse

IMAGE SOURCES



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